IBM COGNOS ANALYTICS

FINAL PROJECT- PHASE 2 (PROJECT PHASE)

COLLEGE CODE: 9632

COLLEGE NAME: PET ENGINEERING COLLEGE

BRANCH: B.TECH

DEPARTMENT: AI & DS

STUDENT NM ID: 1A644D2EAA3488D9AD64A440AFA58DA5

ROLL NO: 963223243022

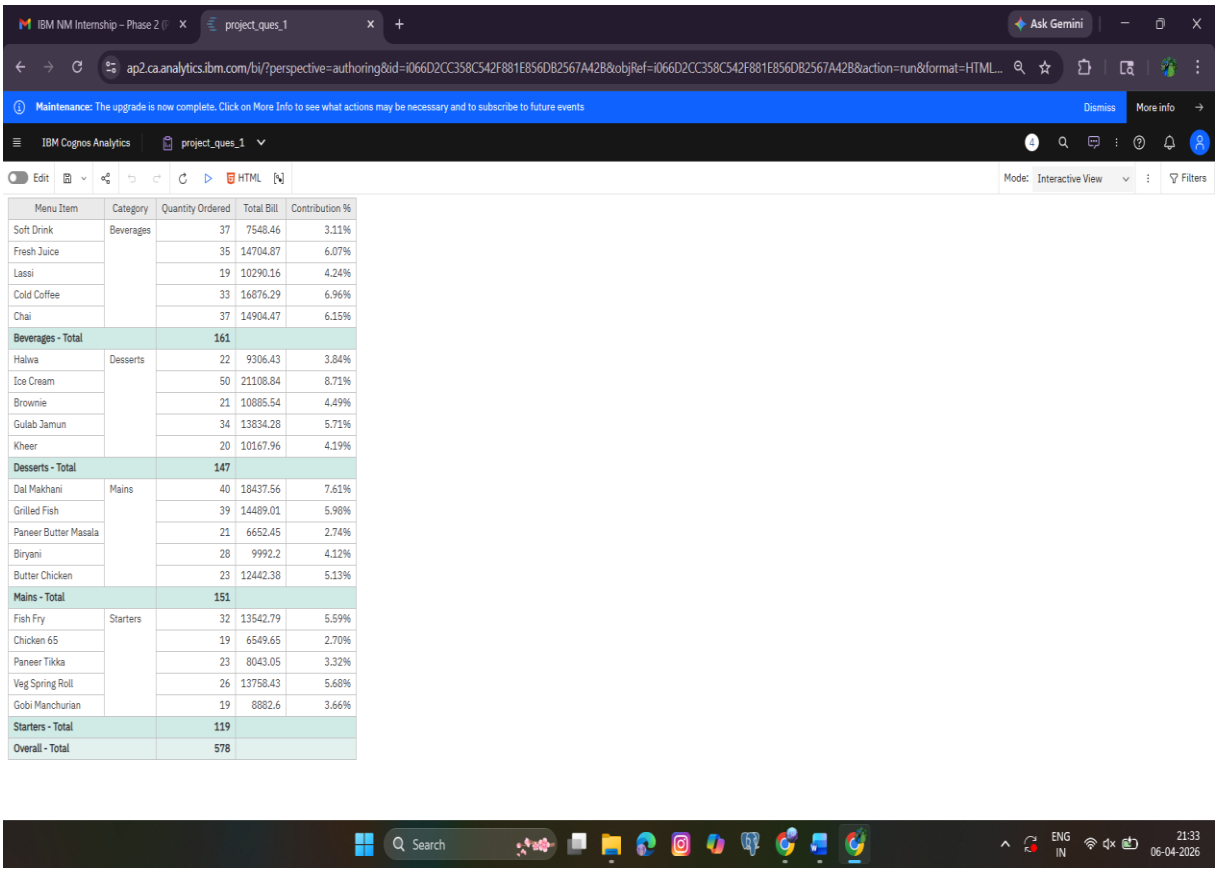
DATE: 22/09/2025

SUBMITTED BY …

NAME : NIRANJAN M

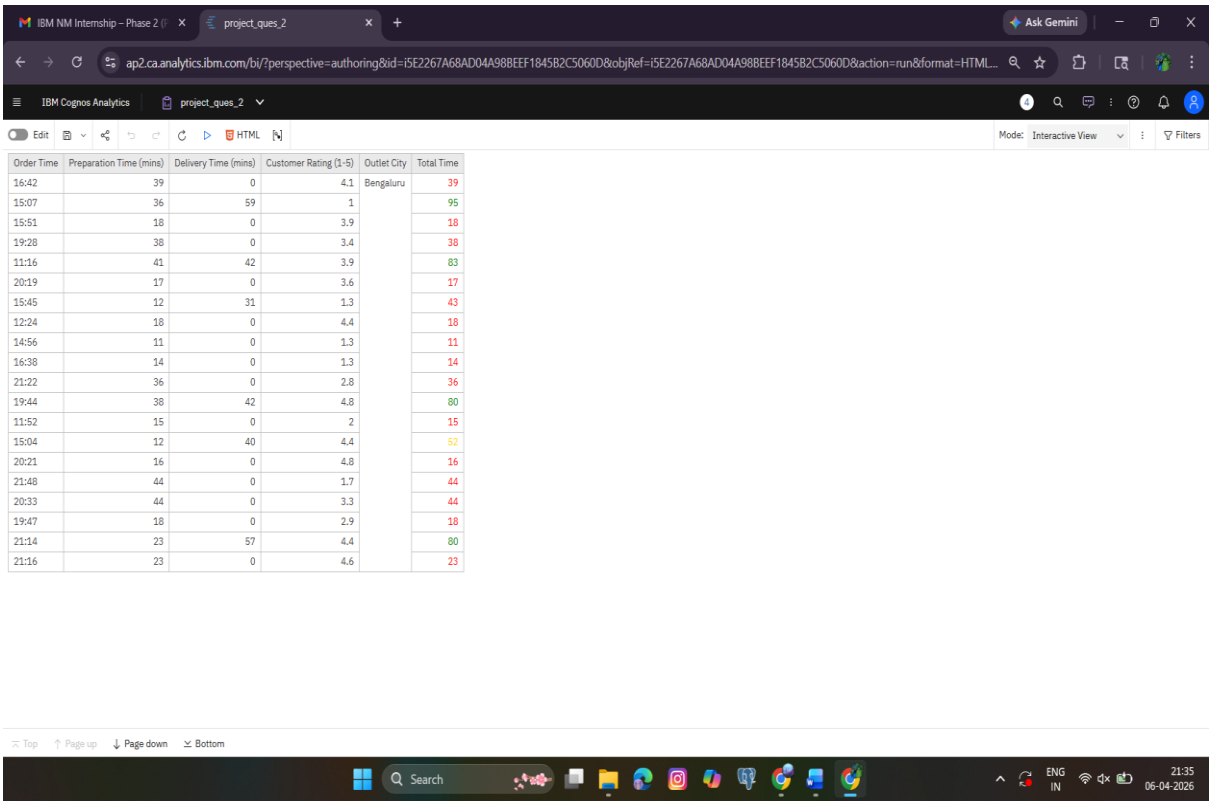
PH NO : 9626417090

Q1 [Reporting] Create a Menu Performance Report showing Menu Item, Category, Total Quantity Ordered, Total Revenue, and Contribution % to Total Revenue. Apply Conditional Styling: Items contributing < 1% to revenue in Red (consider removing). Add prompts for Outlet City and Category. Include Header with SpicePot branding, Footer with page number. Use Palatino Linotype font, size 11.



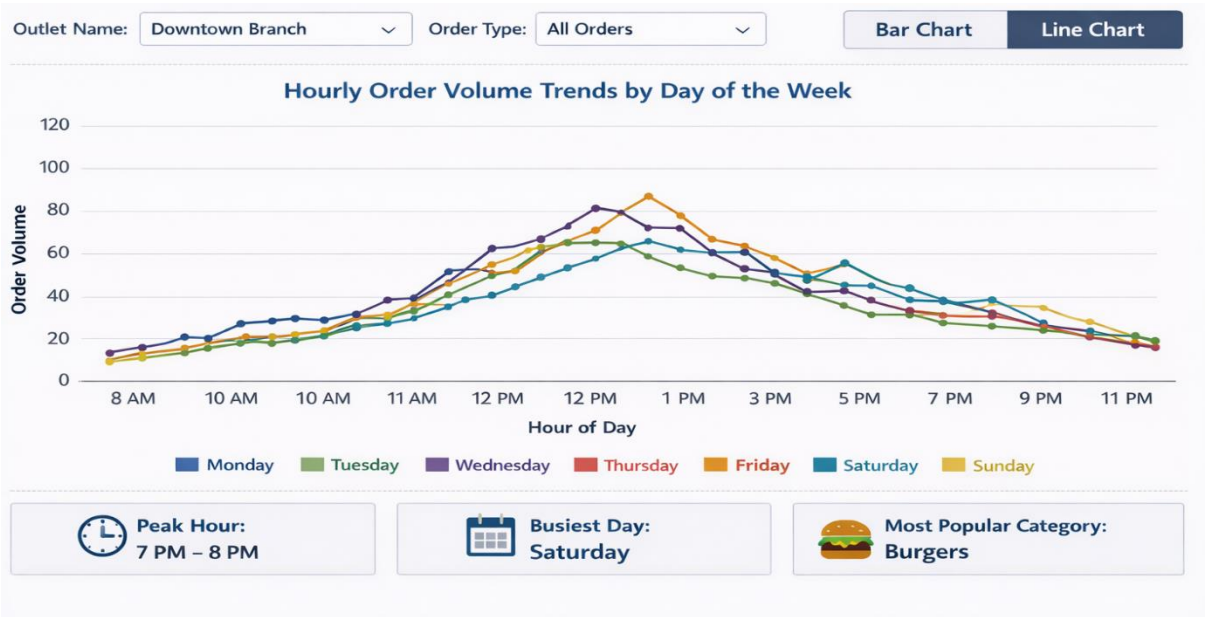
Menu Item	Category	Quantity Ordered	Total Bill	Contribution %
Soft Drink	Beverages	37	7548.46	3.11%
Fresh Juice		35	14704.87	6.07%
Lassi		19	10290.16	4.24%
Cold Coffee		33	16876.29	6.96%
Chai		37	14904.47	6.15%
Beverages - Total		**161**		
Halwa	Desserts	22	9306.43	3.84%
Ice Cream		50	21108.84	8.71%
Brownie		21	10885.54	4.49%
Gulab Jamun		34	13834.28	5.71%
Kheer		20	10167.96	4.19%
Desserts - Total		**147**		
Dal Makhani	Mains	40	18437.56	7.61%
Grilled Fish		39	14489.01	5.98%
Paneer Butter Masala		21	6652.45	2.74%
Biryani		28	9992.2	4.12%
Butter Chicken		23	12442.38	5.13%
Mains - Total		**151**		
Fish Fry	Starters	32	13542.79	5.59%
Chicken 65		19	6549.65	2.70%
Paneer Tikka		23	8043.05	3.32%
Veg Spring Roll		26	13758.43	5.68%
Gobi Manchurian		19	8882.6	3.66%
Starters - Total		**119**		
Overall - Total		**578**		

Q2 [Reporting] Build a Delivery Performance Report listing all delivery orders with Order Time,Preparation Time, Delivery Time, Total Time, and Customer Rating. Add a calculated field: Total Delivery Time = Preparation + Delivery. Apply Conditional Styling: Total Time > 60 mins in Red, 45–60 in Amber, < 45 in Green. Summarize average delivery time by Outlet City in the footer.

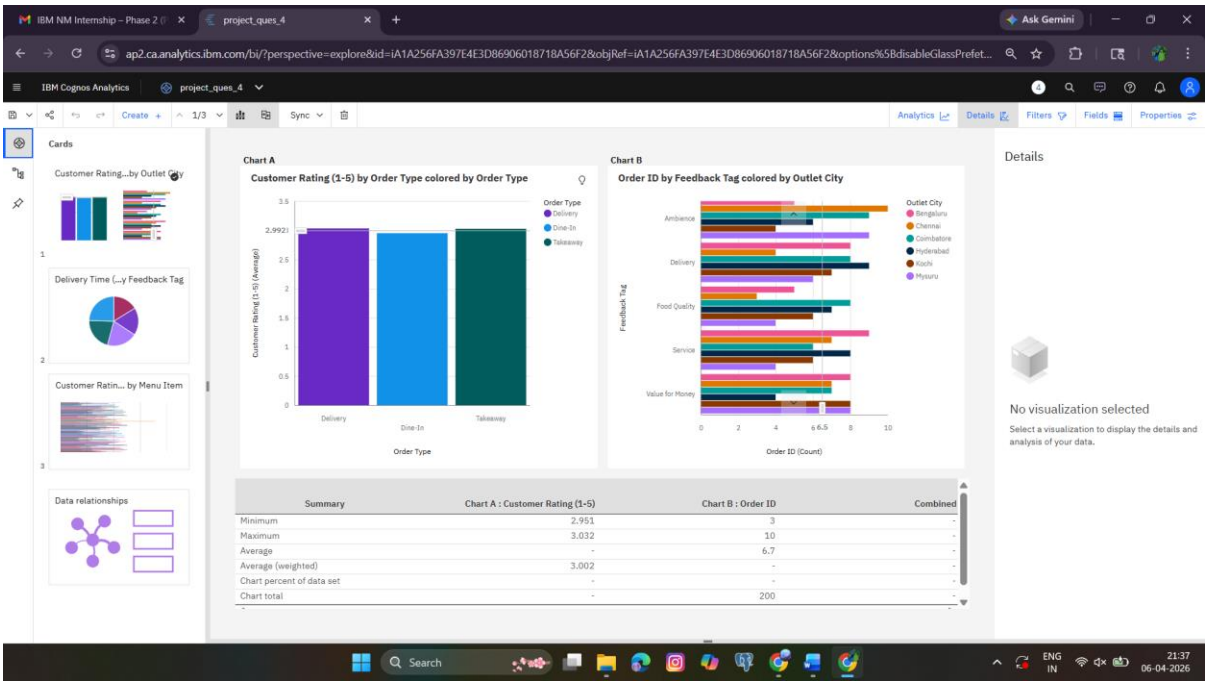


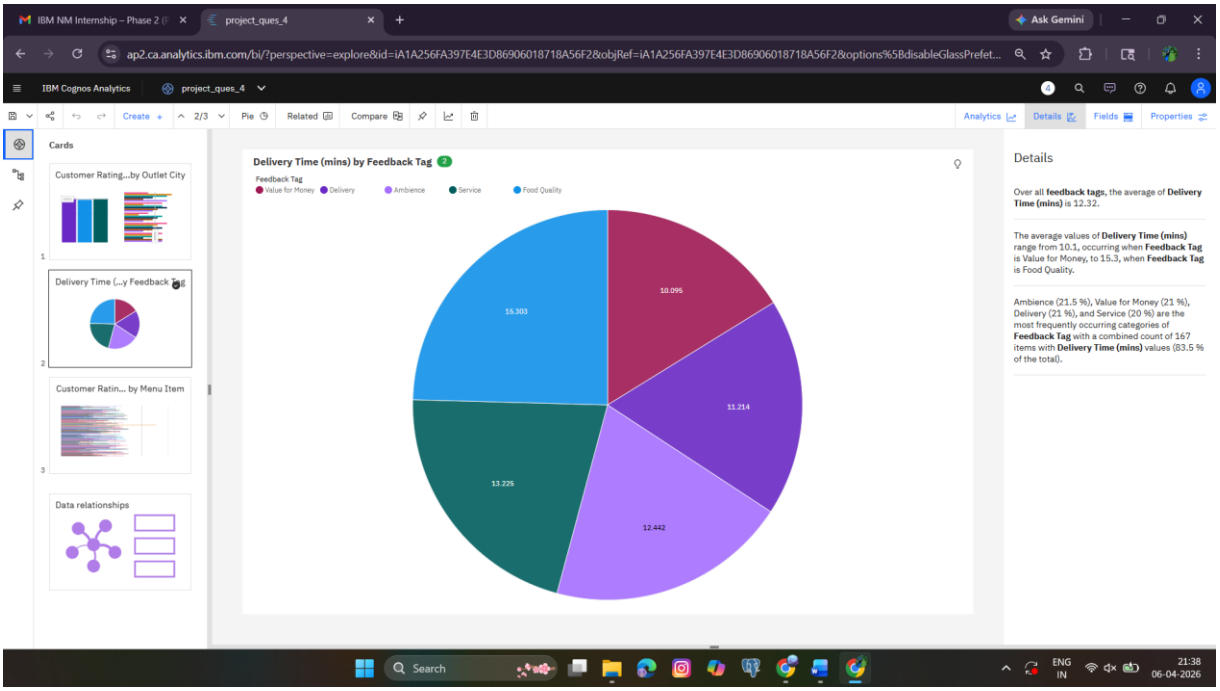
Order Time	Preparation Time (mins)	Delivery Time (mins)	Customer Rating (1-5)	Outlet City	Total Time
16:42	39	0	4.1	Bengaluru	39
15:07	36	59	1		95
15:51	18	0	3.9		18
19:28	38	0	3.4		38
11:16	41	42	3.9		83
20:19	17	0	3.6		17
15:45	12	31	1.3		43
12:24	18	0	4.4		18
14:56	11	0	1.3		11
16:38	14	0	1.3		14
21:22	36	0	2.8		36
19:44	38	42	4.8		80
11:52	15	0	2		15
15:04	12	40	4.4		52
20:21	16	0	4.8		16
21:48	44	0	1.7		44
20:33	44	0	3.3		44
19:47	18	0	2.9		18
21:14	23	57	4.4		80
21:16	23	0	4.6		23

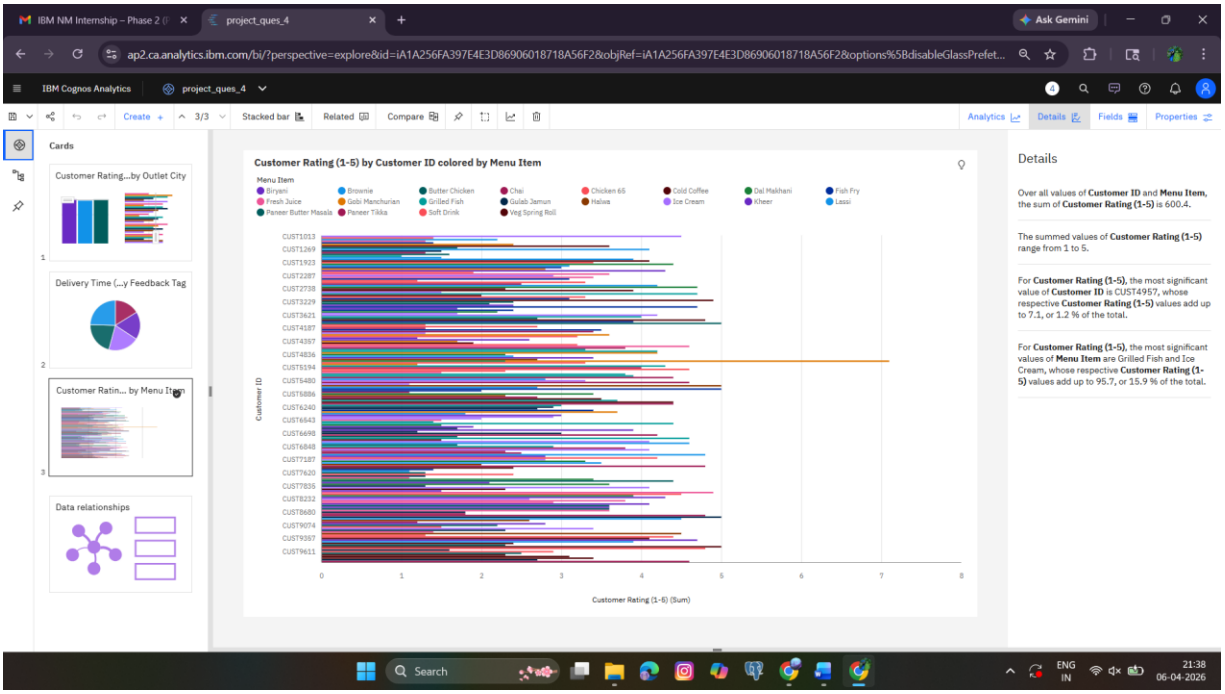
Q3 [Reporting] Using Report Visualize Prompt, allow Outlet Managers to select chart type (Bar or Line) to view Hourly Order Volume Trends across different days of the week. Add prompts for Outlet Name and Order Type. Summarize peak hour, busiest day, and most popular category as text objects. Apply Conditional Styling to highlight hours with Cancelled orders > 10 in Amber. IBM Cognos Analytics - Real-Time Case Study Scenarios



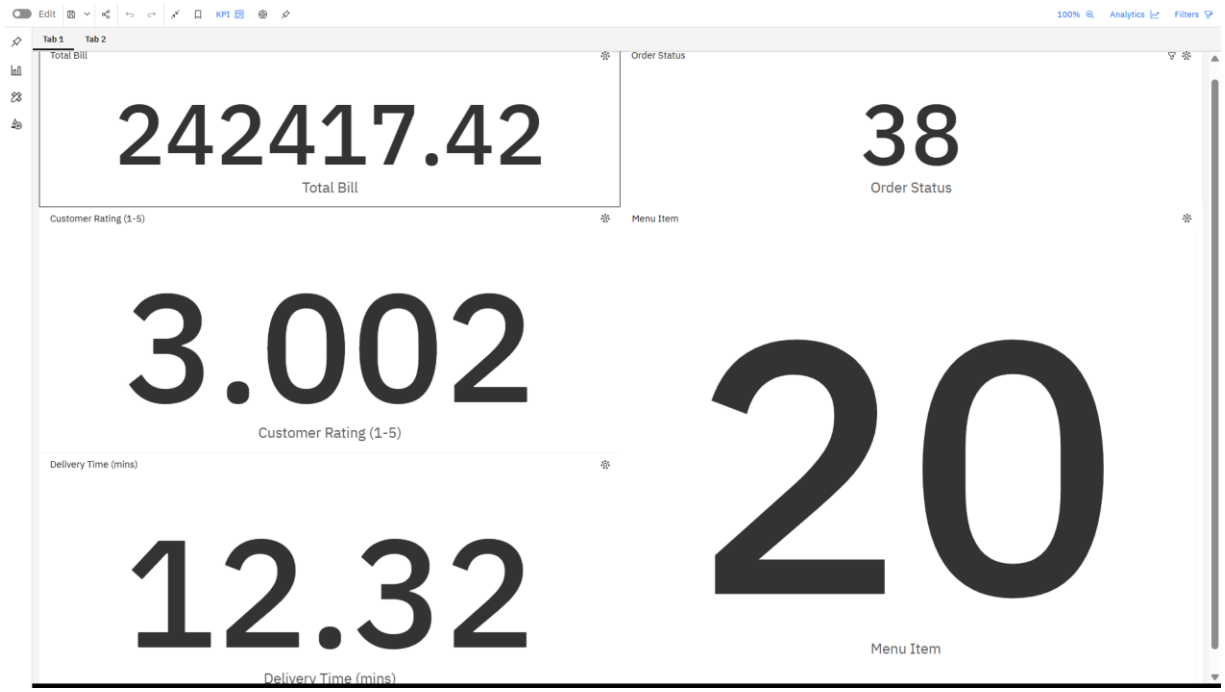
Q4 [Exploration] Use Cognos Exploration to discover patterns in the SpicePot dataset. Explore the relationship between Order Type (Dine-In/Takeaway/Delivery) and Customer Rating. Investigate which Feedback Tags appear most frequently for Delivery orders and how they correlate with Delivery Time. Identify the top 5 best-performing menu items across all outlets and document findings.

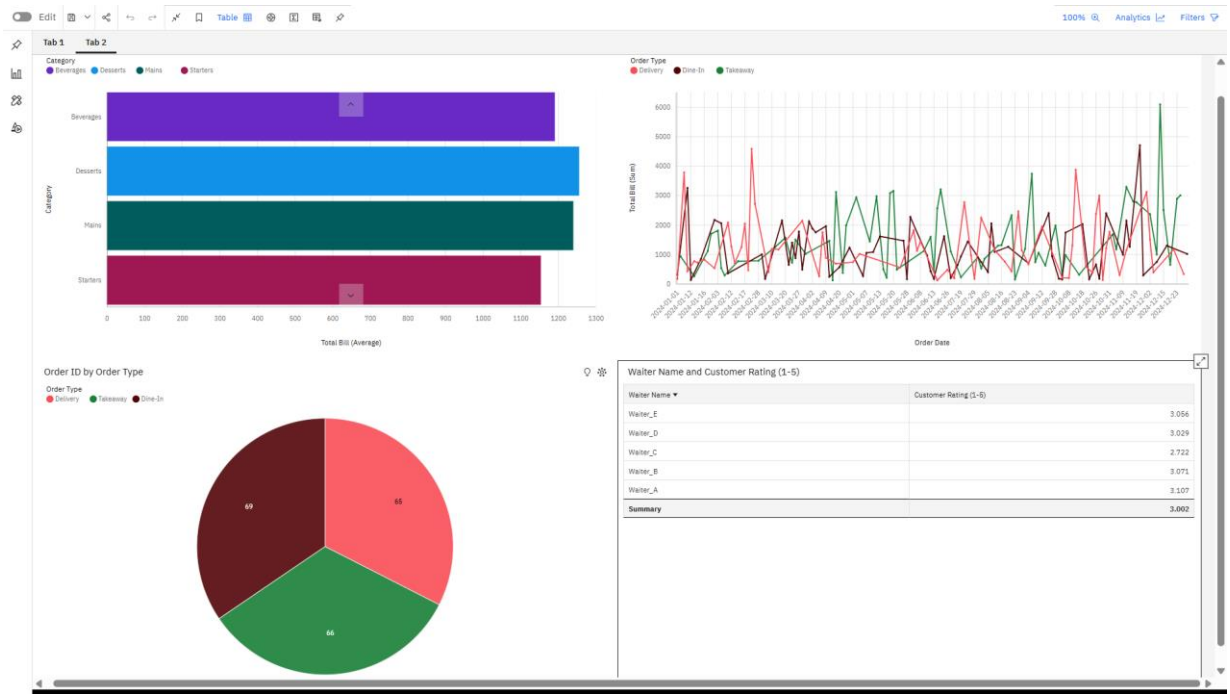




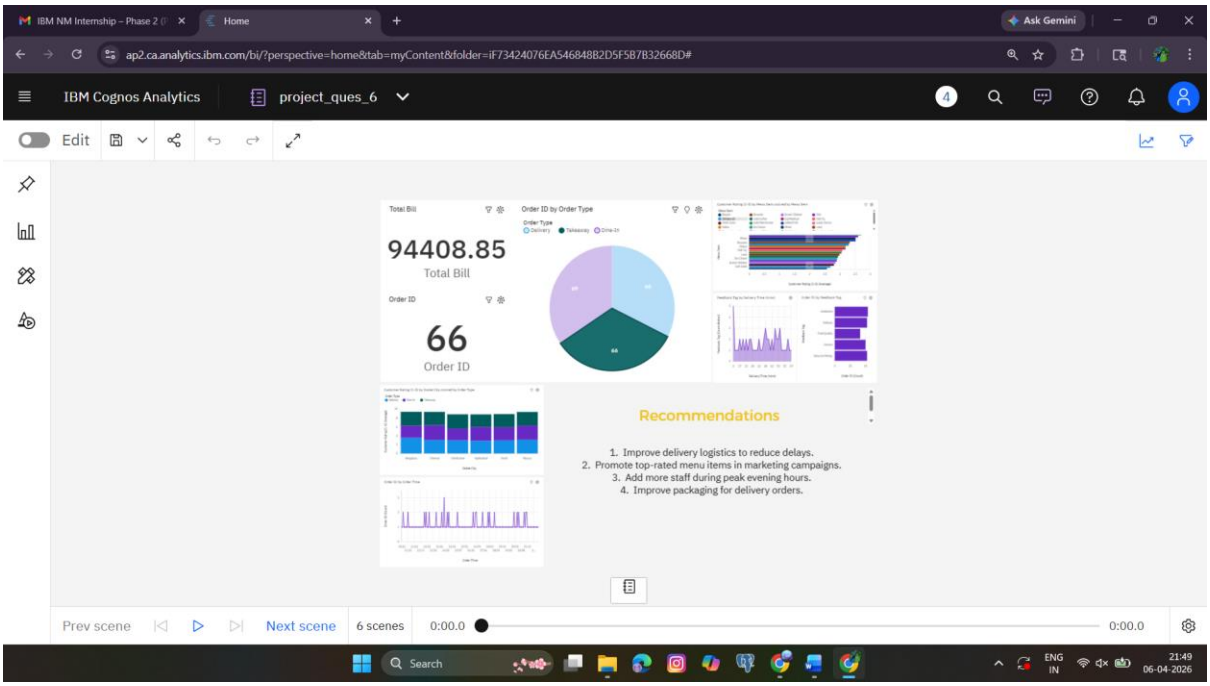

IBM NM Internship – Phase 2
project_ques_4
Ask Gemini
ap2.ca.analytics.ibm.com/bi/?perspective=explore&id=iA1A256FA397E4E3D86906018718A56F2&objRef=iA1A256FA397E4E3D86906018718A56F2&options%5BdisableGlassPrefet...
IBM Cognos Analytics
project_ques_4
Create
3/3
Stacked bar
Related
Compare
Analytics
Details
Fields
Properties
Cards
Customer Rating...by Outlet City
Delivery Time (...y Feedback Tag
Customer Ratin... by Menu Item
Data relationships
Customer Rating (1-5) by Customer ID colored by Menu Item
Menu Item
Biryani
Brownie
Butter Chicken
Chai
Chicken 65
Cold Coffee
Dal Makhani
Fish Fry
Fresh Juice
Gobi Manchurian
Grilled Fish
Gulab Jamun
Halwa
Ice Cream
Kheer
Lassi
Paneer Butter Masala
Paneer Tikka
Soft Drink
Veg Spring Roll
Customer ID
Customer Rating (1-6) (Sum)
Details
Over all values of Customer ID and Menu Item, the sum of Customer Rating (1-5) is 600.4.
The summed values of Customer Rating (1-5) range from 1 to 5.
For Customer Rating (1-5), the most significant value of Customer ID is CUST4957, whose respective Customer Rating (1-5) values add up to 7.1, or 1.2 % of the total.
For Customer Rating (1-5), the most significant values of Menu Item are Grilled Fish and Ice Cream, whose respective Customer Rating (1-5) values add up to 95.7, or 15.9 % of the total.
Search
ENG
IN
21:38
06-04-2026

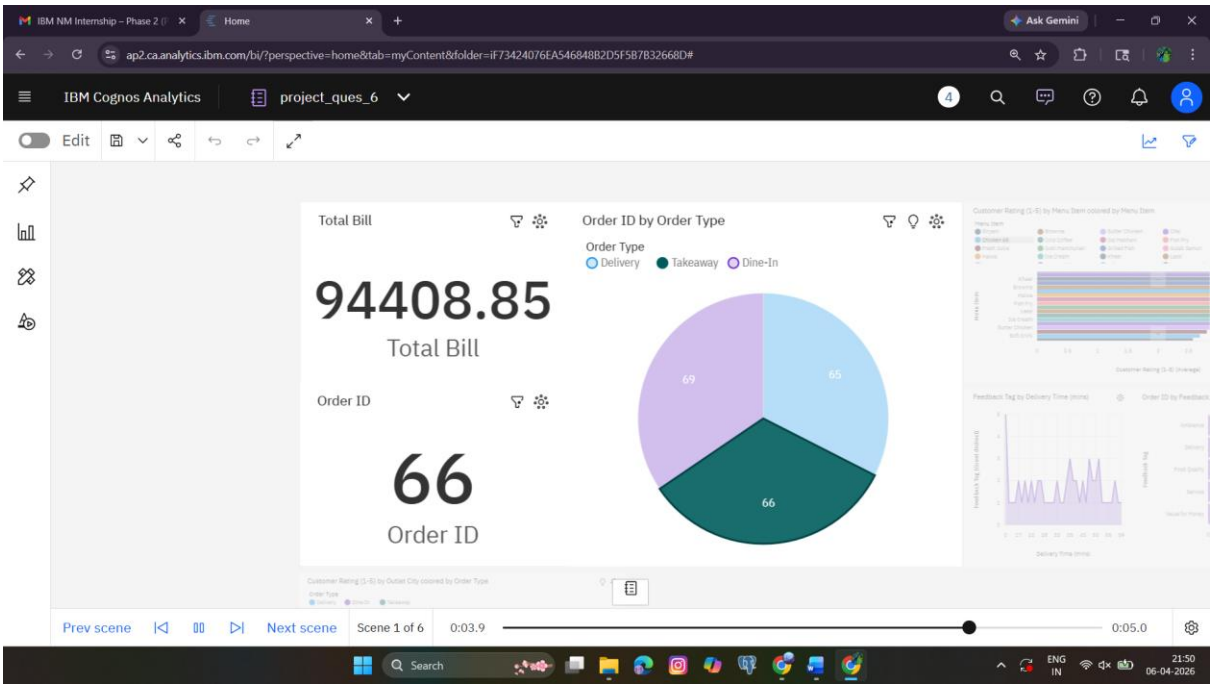
Q5 [Dashboard] Design a SpicePot Operations Dashboard with: (1) KPI tiles — Total Revenue,Average Customer Rating, Order Cancellation Rate, Average Delivery Time, (2) A Bar chart forRevenue by Menu Category, (3) A Line chart for Daily Revenue Trend, (4) A Pie chart for OrderType Distribution (Dine-In/Takeaway/Delivery), (5) A Waiter Performance table by Rating. Add Outlet and Date filters.

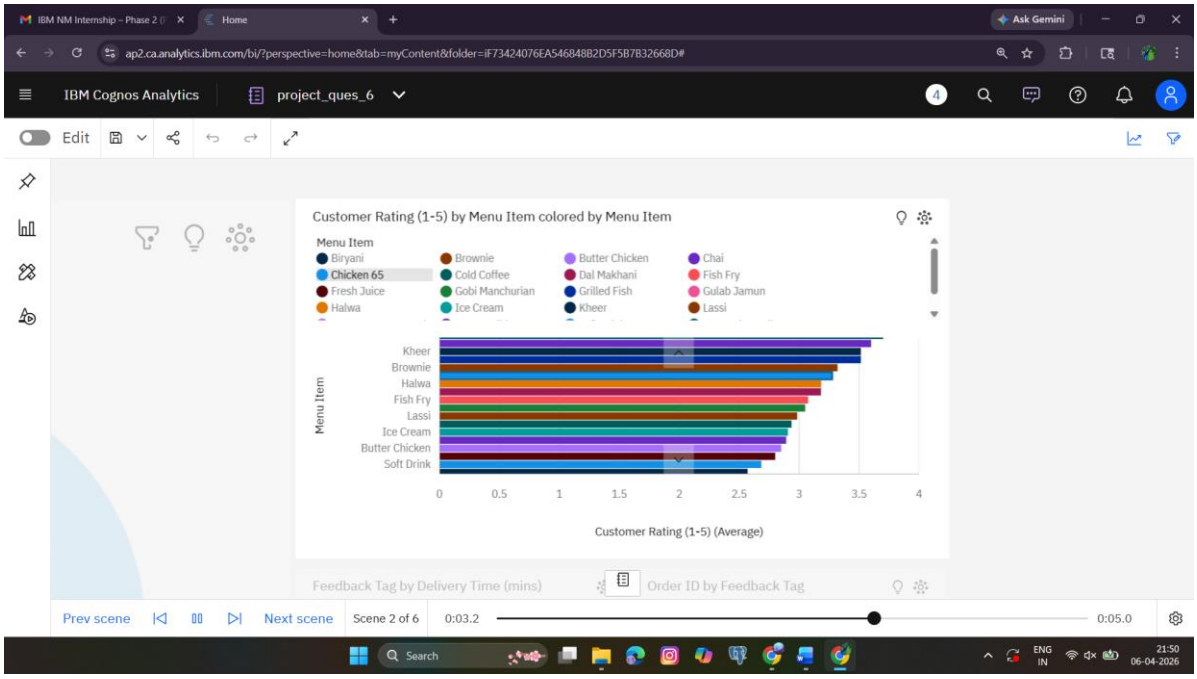


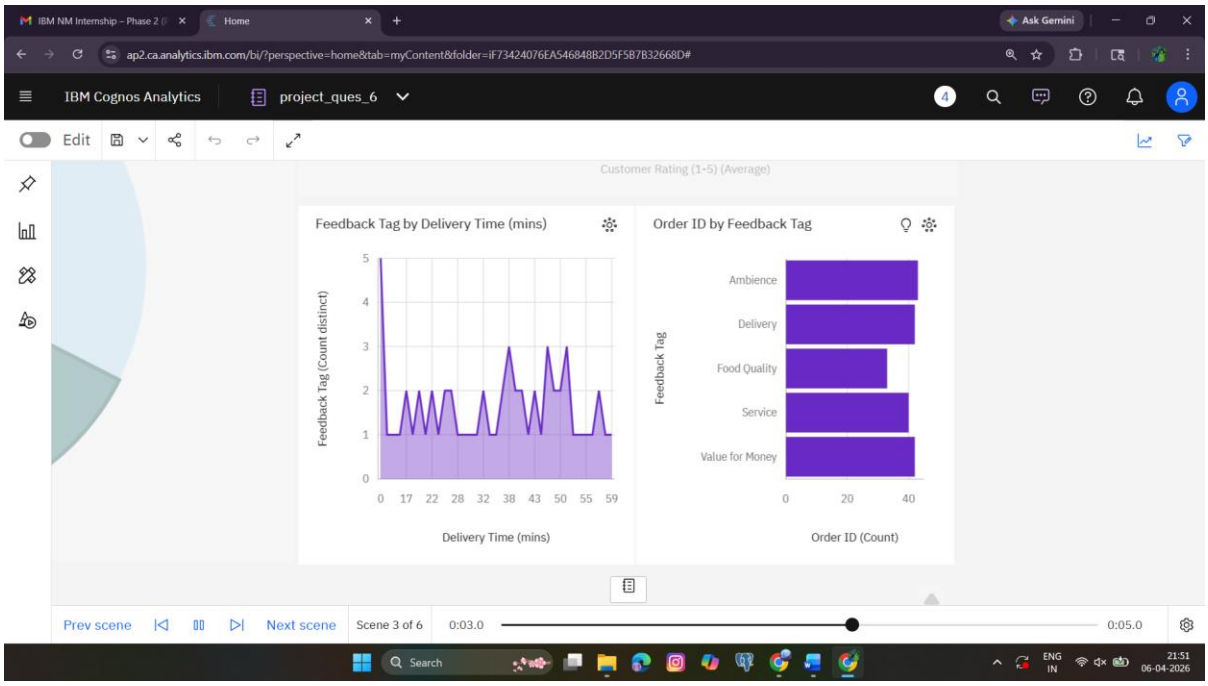


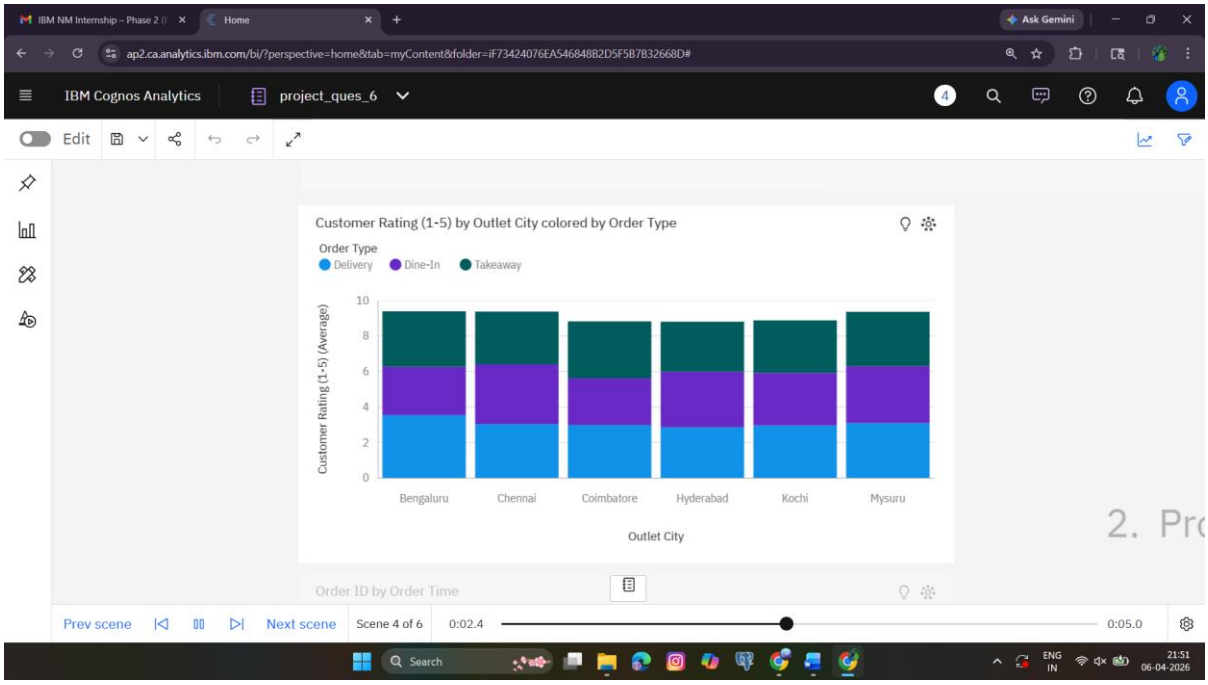
Q6 [Story Making] Create a Cognos Story for SpicePot's Monthly Operations Review: Scene 1 —Chain-wide Sales & Order Overview, Scene 2 — Top & Bottom Performing Menu Items, Scene3 — Delivery Performance & Complaint Analysis, Scene 4 — Customer Satisfaction by Outlet& Order Type, Scene 5 — Peak Hour & Demand Pattern Analysis, Scene 6 — MenuOptimization & Delivery Improvement Recommendations.

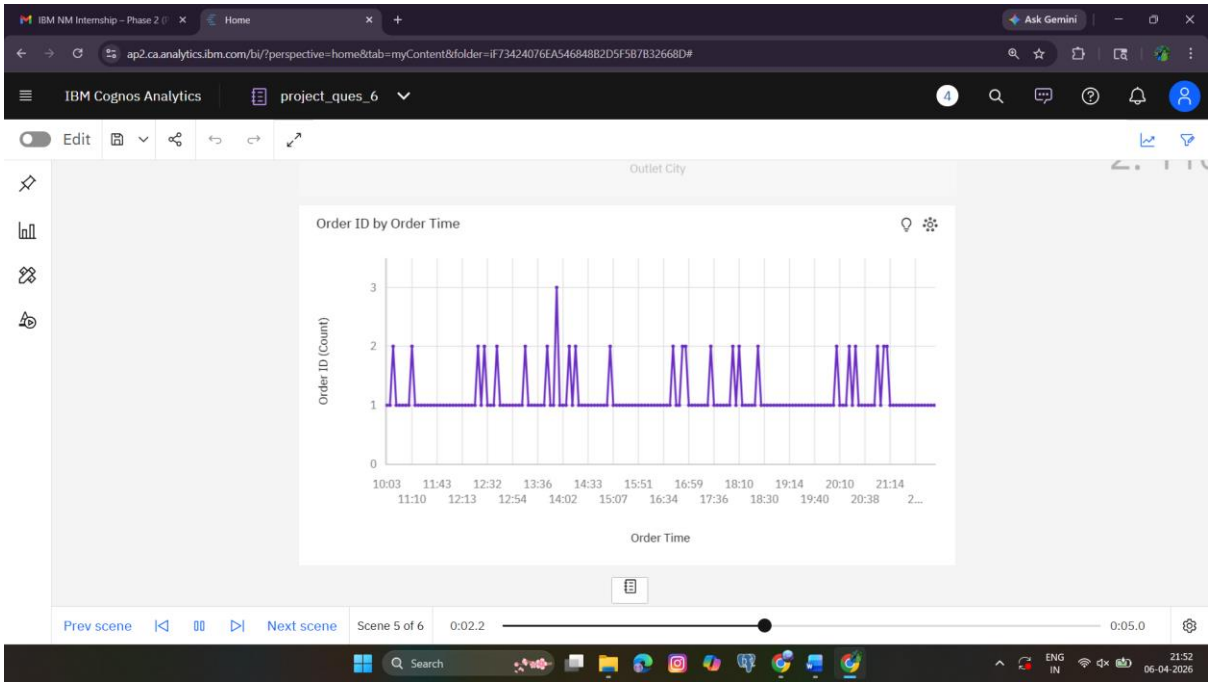


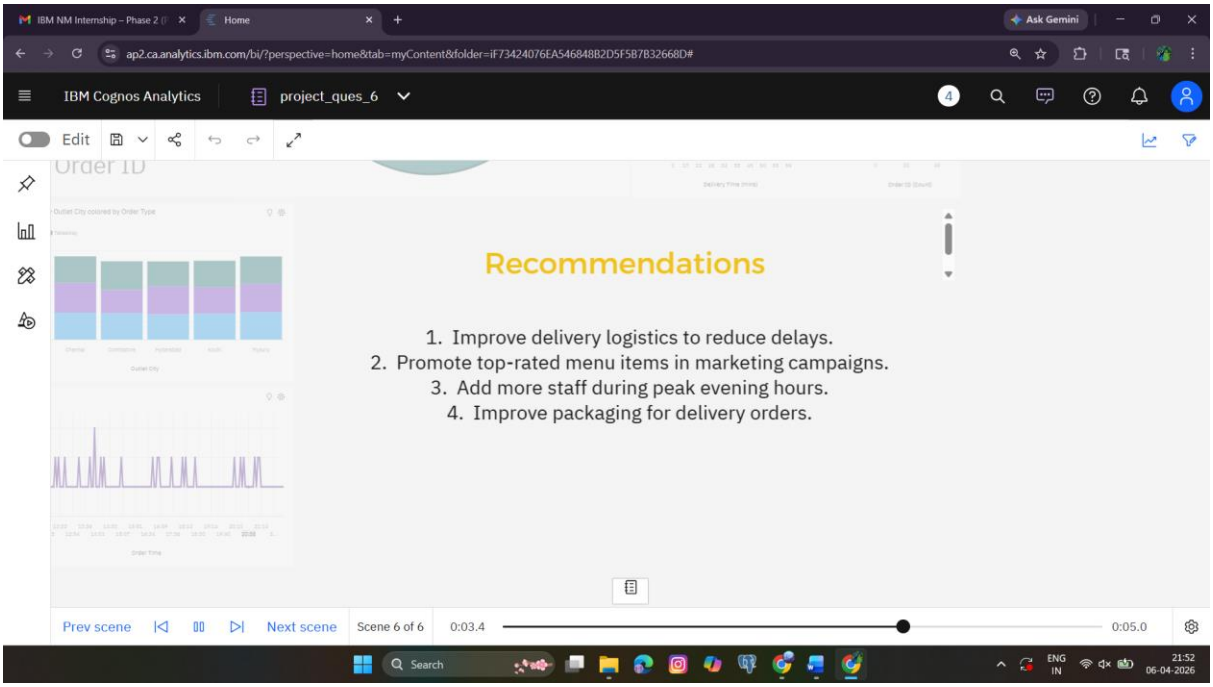



IBM Cognos Analytics
project_ques_6
Customer Rating (1-5) by Menu Item colored by Menu Item
Menu Item
Biryani
Chicken 65
Fresh Juice
Halwa
Brownie
Cold Coffee
Gobi Manchurian
Ice Cream
Butter Chicken
Dal Makhani
Grilled Fish
Kheer
Chai
Fish Fry
Gulab Jamun
Lassi
Kheer
Brownie
Halwa
Fish Fry
Lassi
Ice Cream
Butter Chicken
Soft Drink
Menu Item
0
0.5
1
1.5
2
2.5
3
3.5
4
Customer Rating (1-5) (Average)
Feedback Tag by Delivery Time (mins)
Order ID by Feedback Tag
Prev scene
Next scene
Scene 2 of 6
0:03.2
0:05.0


IBM Cognos Analytics
project_ques_6
Customer Rating (1-5) (Average)
Feedback Tag by Delivery Time (mins)
Feedback Tag (Count distinct)
5
4
3
2
1
0
0
17
22
28
32
38
43
50
55
59
Delivery Time (mins)
Order ID by Feedback Tag
Feedback Tag
Ambience
Delivery
Food Quality
Service
Value for Money
0
20
40
Order ID (Count)
Prev scene
Next scene
Scene 3 of 6
0:03.0
0:05.0


IBM NM Internship – Phase 2
Home
Ask Gemini
ap2.ca.analytics.ibm.com/bi/?perspective=home&tab=myContent&folder=iF73424076EA546848B2D5F5B7B32668D#
IBM Cognos Analytics
project_ques_6
Edit
Customer Rating (1-5) by Outlet City colored by Order Type
Order Type
Delivery
Dine-In
Takeaway
Customer Rating (1-5) (Average)
10
8
6
4
2
0
Bengaluru
Chennai
Coimbatore
Hyderabad
Kochi
Mysuru
Outlet City
Order ID by Order Time
Prev scene
Next scene
Scene 4 of 6
0:02.4
0:05.0
Search
ENG
IN
21:51
06-04-2026


IBM NM Internship – Phase 2
Home
Ask Gemini
ap2.ca.analytics.ibm.com/bi/?perspective=home&tab=myContent&folder=iF73424076EA546848B2D5F5B7B32668D#
IBM Cognos Analytics
project_ques_6
Edit
Outlet City
Order ID by Order Time
Order ID (Count)
3
2
1
0
10:03
11:10
11:43
12:13
12:32
12:54
13:36
14:02
14:33
15:07
15:51
16:34
16:59
17:36
18:10
18:30
19:14
19:40
20:10
20:38
21:14
2...
Order Time
Prev scene
Next scene
Scene 5 of 6
0:02.2
0:05.0
Search
ENG
IN
21:52
06-04-2026


IBM NM Internship – Phase 2
Home
Ask Gemini
ap2.ca.analytics.ibm.com/bi/?perspective=home&tab=myContent&folder=iF73424076EA546848B2D5F5B7B32668D#
IBM Cognos Analytics
project_ques_6
Edit
Recommendations
1. Improve delivery logistics to reduce delays.
2. Promote top-rated menu items in marketing campaigns.
3. Add more staff during peak evening hours.
4. Improve packaging for delivery orders.
Prev scene
Next scene
Scene 6 of 6
0:03.4
0:05.0
Search
ENG
IN
21:52
06-04-2026